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Math 225 Project 3 Fall 2024

10 homework points. Show your work to receive full or partial credit. Your solution should be clear and understandable. This project is to be done in teams of 2 of your choosing, but with a different partner than project 1 and project 2. **This problem is due on Friday, November 15 at the beginning of class.**

Three prehistoric humans, Lug, Uurgh, and Gronk, have carved three rocks in to three shapes: a sphere, a disk, and a ring. Assume a reasonable degree of craftsmanship: the shapes are round enough that they will roll easily, but still rough enough that they do not slip on the ground. Also assume that each rock has the same radius and the same density throughout. Lug, Uurgh, and Gronk play an exciting game of letting the objects roll down a hill to see which one gets to the bottom first. They repeat the game several times, but the outcome is always the same. Your goal in this project is to use some physics and math to give a complete mathematical argument of why this is so. You will assume that the disk and the ring have the same thickness T and that the radius of the inner ring is i units.

1. Use triple integrals and spherical coordinates to compute the moment of inertia of the sphere about an axis that runs through the center of the sphere. Also list the the mass of the sphere.

$$Outer\ Radius = r = \rho, Density: \rho(x,y,z) = \rho_{de} \text{ (to distinguish from } \rho\text{)}$$

$$Mass = \iiint_S dV = \rho_{de} \int_0^{2\pi} \int_0^{\pi} \int_0^{\rho} \rho^2\, sin(\phi)\, d\rho d\phi d\theta = \rho_{de} \int_0^{2\pi} \int_0^{\pi} (\frac{\rho^3}{3} sin(\phi)) \Big|_0^{\rho} d\phi d\theta$$

$$M = \frac{\rho_{de}\rho^3}{3} \int_0^{2\pi} \int_0^{\pi} sin(\phi)\, d\phi d\theta = \frac{\rho_{de}\rho^3}{3} \int_0^{2\pi} (-cos(\phi)\, d\theta) \Big|_0^{\pi} d\theta$$

$$M = -\frac{\rho_{de}\rho^3}{3} \int_0^{2\pi} (-1-1) d\theta = \frac{2\rho_{de}\rho^3}{3} \int_0^{2\pi} d\theta = \frac{2\rho_{de}\rho^3}{3} (\theta) \Big|_0^{2\pi} = \frac{4\pi\rho_{de}\rho^3}{3},$$

$$where\ \rho_{de} = density, and\ \rho = radius$$

$$M = \frac{4\pi\rho_{de}\rho^3}{3}$$

$$x^2 + y^2 = r^2 = (\rho \sin(\phi))^2 = \rho^2 \sin^2(\phi)$$

$$I_z = \iiint_S (x^2 + y^2)\rho(x,y,z) dV = \rho_{de} \iiint_S (\rho^2 \sin^2(\phi))\rho^2 \sin(\phi)\, d\rho d\phi d\theta$$

$$I_z = \rho_{de}\int_0^{2\pi}\int_0^{\pi}\int_0^{\rho}\rho^4\sin^3(\phi)\,d\rho d\phi d\theta = \frac{\rho_{de}}{5}\int_0^{2\pi}\int_0^{\pi}(\rho^5\sin^3(\phi))_0^{\rho}d\phi d\theta$$

$$I_z = \frac{\rho_{de}\rho^5}{5}\int_0^{2\pi}\int_0^{\pi}\sin^3(\phi)\,d\phi d\theta = \frac{\rho_{de}\rho^5}{5}\int_0^{2\pi}\int_0^{\pi}\sin(\phi)\,(1-\cos^2(\phi))d\phi d\theta$$

$$u = \cos(\phi)\,, du = -\sin(\phi)\,, u_1 = \cos(0) = 1, u_2 = \cos(\pi) = -1$$

$$I_z = -\frac{\rho_{de}\rho^5}{5}\int_0^{2\pi}\int_1^{-1}(1-u^2)dud\theta = \frac{\rho_{de}\rho^5}{5}\int_0^{2\pi}\int_{-1}^{1}(1-u^2)dud\theta$$

$$I_z = \frac{\rho_{de}\rho^5}{5}\int_0^{2\pi}\left(u-\frac{1}{3}u^3\right)\begin{matrix}1\\-1\end{matrix}d\theta = \frac{\rho_{de}\rho^5}{5}\int_0^{2\pi}\left(1-\frac{1}{3}+1-\frac{1}{3}\right)d\theta$$

$$I_z = \frac{\rho_{de}\rho^5}{5}\left(\frac{4}{3}\right)\int_0^{2\pi}d\theta = \frac{4\rho_{de}\rho^5}{15}(\theta)\begin{matrix}2\pi\\0\end{matrix} = \frac{4\rho_{de}\rho^5}{15}(2\pi) = \frac{8\pi\rho_{de}\rho^5}{15}$$

By substituting the sphere's mass into its inertia equation, we eliminated the density term and found the inertia in terms of mass and radius.

$$M = \frac{4\pi\rho_{de}\rho^3}{3}, \qquad \rho_{de} = \frac{3M}{4\pi\rho^3}$$

$$I_x = I_y = I_z = \frac{8\pi\rho_{de}\rho^5}{15} = \frac{8\pi\rho^5}{15}\left(\frac{3M}{4\pi\rho^3}\right) = \frac{2}{5}M\rho^2 = \frac{2}{5}Mr^2, where\ M = mass, r = radius$$

$$I = \frac{2}{5}Mr^2$$

2. Use triple integrals and cylindrical coordinates to compute the moment of inertia of the disk about an axis that runs through the center of the disk normal to the circular shape. Also list the mass of the disk.

$$m_{disk} = \int_0^{2\pi}\int_0^{r}\int_0^{T} \rho(x,y,z) r \, dz dr d\theta = \rho_{de}\int_0^{2\pi}\int_0^{r} Tr \, dr d\theta$$

$$= \rho_{de} T \int_0^{2\pi} \frac{1}{2} r^2 \, d\theta = r^2 \rho_{de} T\pi = m_{disk}$$

$$I_Z = \iiint_S (x^2+y^2)\rho(x,y,z) dV = \int_0^{2\pi}\int_0^{r}\int_0^{T} (r^2)\, \rho_{de} r \, dz dr d\theta$$

$$= \rho_{de}\int_0^{2\pi}\int_0^{r}\int_0^{T} r^3 \, dz dr d\theta = \rho_{de}\int_0^{2\pi}\int_0^{r} r^3 T \, dr d\theta = \rho_{de}\int_0^{2\pi} \frac{1}{4} r^4 T \, d\theta$$

$$= \frac{\rho_{de}}{2} r^4 T\pi$$

$$M = r^2 \rho_{de} T\pi, \qquad \rho_{de} = \frac{M}{r^2 T\pi}$$

$$I_Z = \frac{1}{2} r^4 T\pi \left(\frac{1}{r^2 M \, T\pi}\right) = \frac{1}{2} M r^2 = I_Z$$

3. Use triple integrals and cylindrical coordinates to compute the moment of inertia of the ring about an axis that runs through the center of the ring normal to the circular shape. Also list the mass of the ring.

$$I_{z_{ring}} = \int_0^{2\pi}\int_i^r\int_0^T (x^2+y^2)\rho(x,y,z)dV = \int_0^{2\pi}\int_i^r\int_0^T (r^2)\rho_{de} r\, dzdrd\theta$$
$$= \rho_{de}\int_0^{2\pi}\int_i^r Tr^3 drd\theta = \rho_{de}T\int_0^{2\pi}\frac{1}{4}(r^4-i^4)d\theta$$
$$= \rho_{de}\int_0^{2\pi}\int_i^r\int_0^T r^3 dzdrd\theta = \frac{\rho_{de}T}{4}(r^4-i^4)(2\pi) = \frac{\rho_{de}T\pi}{2}(r^4-i^4)$$

$$m_{ring} = \int_0^{2\pi}\int_i^r\int_0^T \rho(x,y,z)r\, dzdrd\theta = \rho_{de}\int_0^{2\pi}\int_i^r Tr\, drd\theta$$
$$= \rho_{de}T\int_0^{2\pi}\frac{1}{2}(r^2-i^2)\ d\theta = \rho_{de}T(r^2-i^2)\,\pi = m_{ring}$$

$$M = \rho_{de}T(r^2-i^2)\pi, \qquad \rho_{de} = \frac{M}{T(r^2-i^2)\pi}$$
$$I_Z = \frac{T\pi}{2}(r^4-i^4)\frac{M}{T(r^2-i^2)\pi} = \frac{1}{2}\frac{M\,(r^2+i^2)(r^2-i^2)}{(r^2-i^2)} = \frac{1}{2}M\,(r^2+i^2) = I_Z$$

4. When rolling without slipping, the acceleration a of an object down an inline making an angle of θ with the horizontal is $a = \dfrac{g\sin(\theta)}{1+\frac{I}{mr^2}}$ where g is the acceleration due to gravity, I is the moment of inertia, m is the mass of the object, and r is the distance from the axis of rotation to the ground (radius). Find the acceleration for each of the three objects and conclude the order they will reach the bottom of the hill in.

$$a = \frac{gsin(\theta)}{1+\frac{I}{Mr^2}}$$

$$a_{sph} = \frac{gsin(\theta)}{1+\frac{\frac{2}{5}Mr^2}{Mr^2}} = \frac{gsin(\theta)}{1+\frac{2}{5}} = \frac{gsin(\theta)}{\frac{7}{5}} = \frac{5}{7}gsin(\theta) = a_{sph}$$

$$a_{disc} = \frac{gsin(\theta)}{1 + \frac{\frac{1}{2}Mr^2}{Mr^2}} = \frac{gsin(\theta)}{1 + \frac{1}{2}} = \frac{gsin(\theta)}{\frac{3}{2}} = \frac{2}{3}gsin(\theta) = a_{disc}$$

$$a_{ring} = \frac{gsin(\theta)}{1 + \frac{\frac{1}{2}M(r^2 + i^2)}{Mr^2}} = \frac{gsin(\theta)}{1 + \frac{1}{2}\left(1 + \frac{i^2}{r^2}\right)} = \frac{1}{\frac{3}{2} + \frac{i^2}{2r^2}}gsin(\theta)$$

The ring's acceleration is variable, so we decided to evaluate it at i = r.

$$a_{ring}(i = r) = \frac{1}{\frac{3}{2} + \frac{r^2}{2r^2}}gsin(\theta) = \frac{1}{\frac{3}{2} + \frac{1}{2}}gsin(\theta) = \frac{1}{2}gsin(\theta) = a_{ring}(i = r)$$

The three objects will likely reach the ground in this order (first to last): sphere, disc, ring. This is because $\frac{5}{7} > \frac{2}{3} > \frac{1}{2}$.

5. If the stone in one of the shapes is denser than in the others, will this change the results? If the radius of one of the shapes is larger than the others, will this change the results? Does the thickness of the ring and disk make a difference?

   Neither the density nor radius terms affect the final answers because they get canceled out in the process. Likewise, the thickness term T cancels out, so it also does not make a difference.